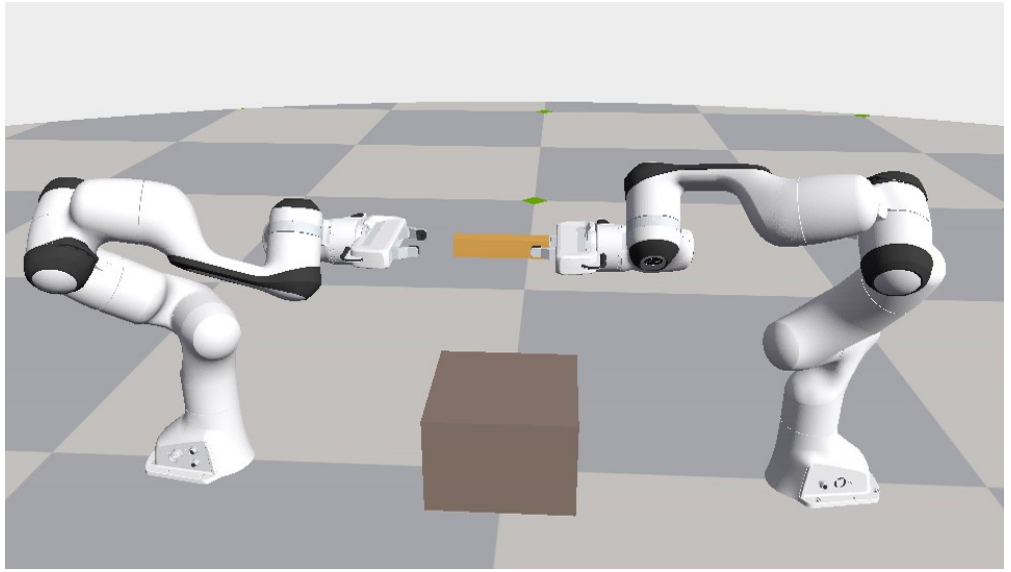
Robust Bimanual Handover — v2 Follow-up Study

Geometric coordination vs residual RL · Isaac Lab 3.0 / Isaac Sim 6.2 (PhysX) · CSCI 645, Fall 2026 · Yan Wen

- Two Franka Panda arms, parallel-jaw grippers, one rigid bar; success = 2 s receiver-only hold
- v1: geometry module decisive, but the selector picked untrained policies (8/9 runs)
- v2: audit v1 → diagnose residuals → ablate geometry → preregistered harder experiment
- Answer: strong geometry suffices; learned residuals on top add nothing and lose to a tuned constant



Task and methods

method	receiver target	release
nominal	fixed approach–grasp schedule	scheduled
geometry	+ alignment (A) + shared-holding anchor (H)	contact-gated (G)
residual	+ learned 3-D residual, 2 cm bound (PPO)	scheduled
combined	geometry on top of nominal + residual	contact-gated
combined_no_phase	combined without 4 contact-phase inputs	contact-gated
constant controls	residual replaced by a fixed offset	as base

- Perturbations: bar start position (per axis) and release time
- 20 Hz control through differential IK, fixed orientations
- PPO (RSL-RL 5.5.1), 1,024 envs, 10M transitions per run
- Same evaluator and case realizations for every controller

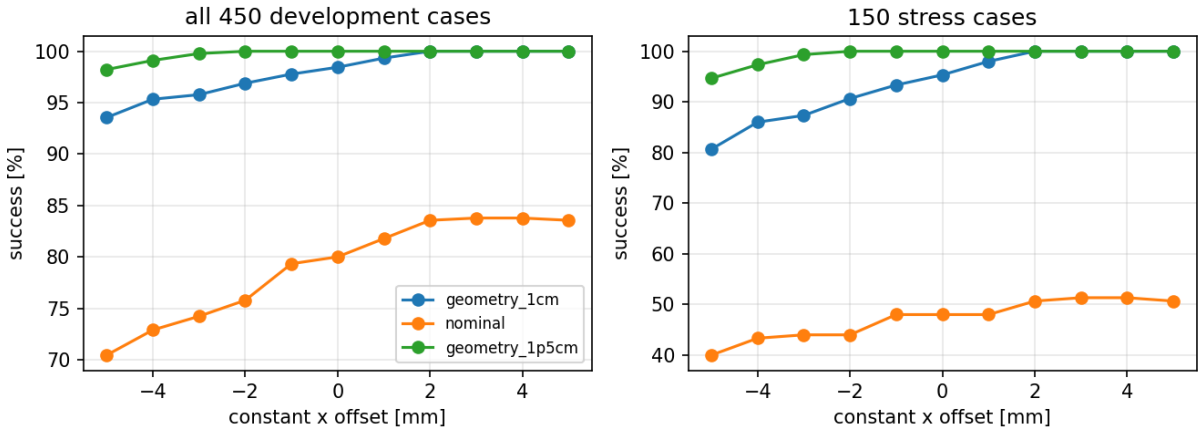
Stage 1 — Audit of v1

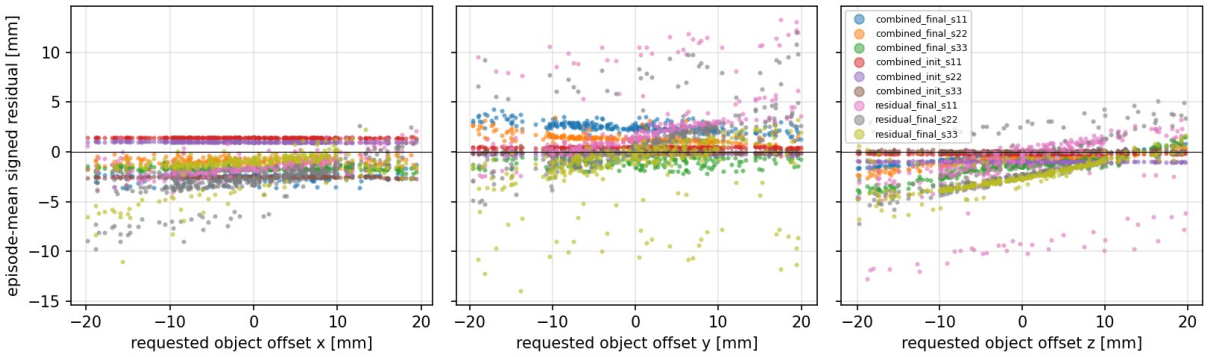
reproducible; two labelling/logging errors, one interpretation gap

- Fresh download: 21/21 checksums, 556 archive members re-hashed; tables regenerate byte for byte
- 19,800 + 16,200 records, identical case IDs, 0 duplicates / invalid resets / errors
- E1: aggregate labels '1,400 in-range / 400 stress' → 1,200 non-stress / 600 stress (numbers unchanged)
- E2: recorded episode_return omits the terminal-step reward (records written before rewards) — logging only
- E3: 'untrained' step-0 policies output a seed-specific constant (+1.4 / +1.0 / −2.5 mm), not zero
- Replay of 23 predeclared cases × 20 policies in new processes: 456/460 and 457/460 outcomes agree
  - all disagreements are failure-boundary cases; GPU physics is not bitwise reproducible across batch layouts
- Negative controls physical: table support → receiver grips 17 N, giver open, table 1.3–1.6 N → 0/20 success

Stage 2 — What do the learned residuals do?

450 development cases, archived v1 checkpoints

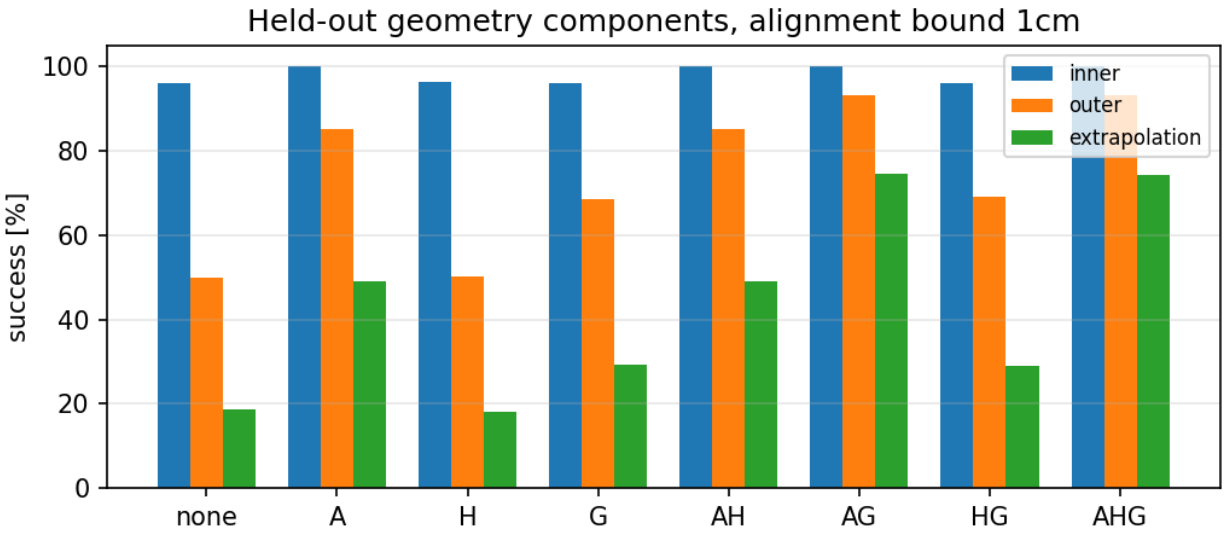


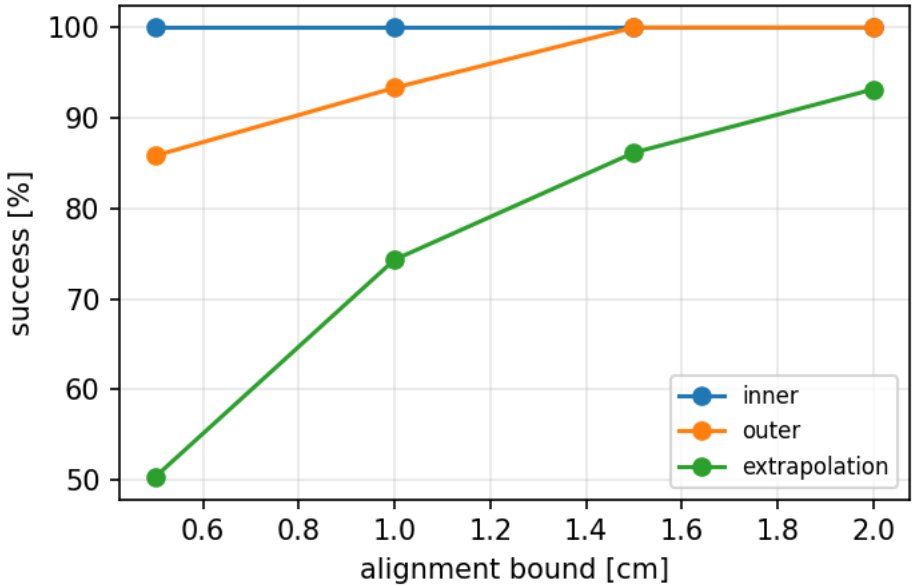


- Geometry + constant +3 mm: 450/450 (geometry + 0: 443)
- Trained combined: near-constant −1…−2.5 mm x bias (spread ≤ 0.4 mm), the harmful direction
  - −7.8 pp vs the constant on stress cases [−13.1, −3.3]
- Failures = palm hits bar end when the object is shifted +15…+19 mm toward the receiver
- Residual-only policies are state dependent (slope 0.07–0.16) but ≈ best constant (+4.4 [−2.0, +11.3])

Stage 3 — Which geometry component matters?

held-out v2 suite, residual = 0





- Alignment A: +18.9 pp (1 cm), +24.0 pp (1.5 cm)
- Release gate G: +7.8 / +7.3 pp; up to +25.5 pp beyond the alignment bound
- Shared-holding anchor H: no measurable effect (±0.1 pp)
- Bound 0.5 / 1 / 1.5 / 2 cm → 84.0 / 91.9 / 96.5 / 98.3 % (all cases)

Stage 4 — Preregistered v2 protocol

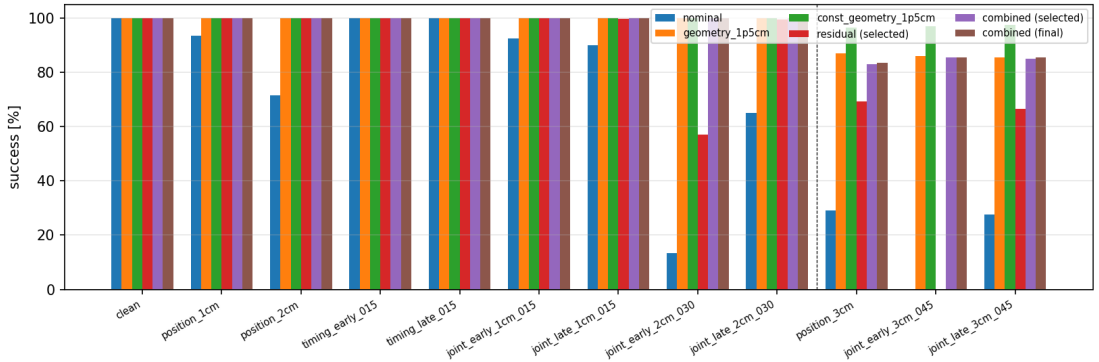
frozen at tag experiment-v2 (protocol sha256 bd79ae51…)

- Training range widened to ±2 cm / ±0.30 s (curriculum); 1.5 cm geometry in combined methods
- Zero-mean actor initialization (pilot P2 chosen by a predeclared rule; 14.75M of 15M pilot budget)
- Selection excludes step 0; ties → fewer collisions → smaller residual → latest
- Constants tuned on 750 development cases: nominal +9 mm, geometry 1 cm +7 mm, geometry 1.5 cm +10 mm
- Final suite: 12 conditions × 200 fresh cases incl. ±3 cm extrapolation (feasibility-checked)
- Disclosed: the training range is already solved by 1.5 cm geometry; validation near-saturated
- 9 runs × 10,027,008 transitions; 117,600 evaluation episodes, all reconciled (0 invalid, 0 errors)

Stage 5 — Held-out results

success %, 2,400 fresh cases; learned = mean of 3 seeds (selected checkpoints)

controller	inner	outer ±2 cm	extrap. ±3 cm	all
nominal	96.0	50.0	18.8	65.2
nominal + 9 mm	99.9	57.0	20.3	69.3
geometry 1.5 cm	100.0	100.0	86.2	96.5
geometry 1.5 cm + 10 mm	**100.0**	**100.0**	**97.0**	**99.2**
residual (PPO)	99.9	85.5	45.2	82.6
combined (PPO)	100.0	100.0	84.4	96.1
combined_no_phase	100.0	100.0	85.2	96.3



- Everything with 1.5 cm geometry is 100 % inside the training range
- Differences appear only in extrapolation

Primary comparisons

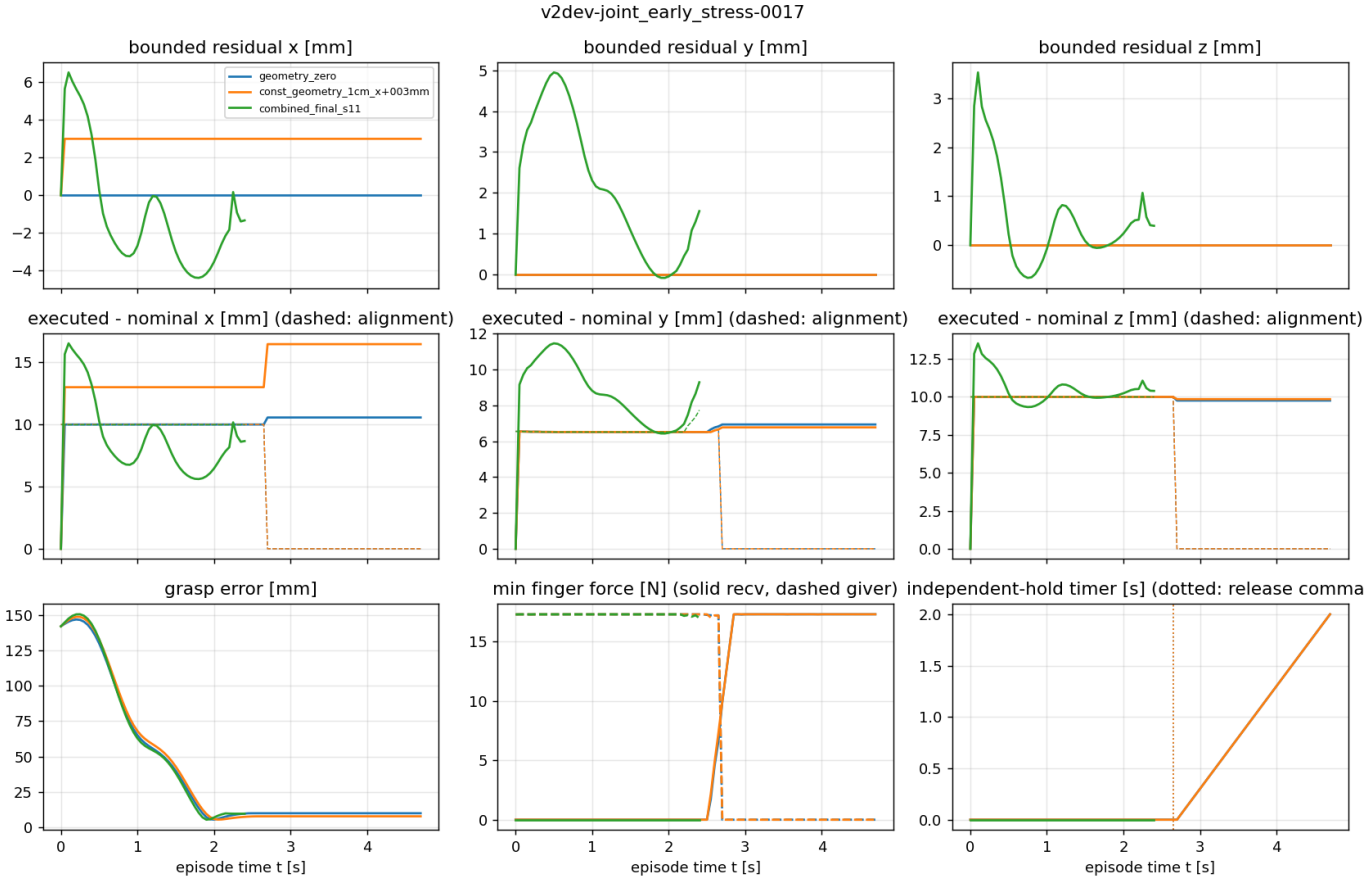
paired seed × case bootstrap, pp [95% CI], all 2,400 cases

comparison	selected	final checkpoints
combined − geometry 1.5 cm	-0.4 [-0.8, -0.2]	-0.4 [-0.6, -0.1]
combined − (geometry + 10 mm)	-3.2 [-4.0, -2.3]	-3.1 [-3.9, -2.2]
residual − nominal	+17.4 [+15.9, +19.0]	+17.7 [+16.1, +19.2]
residual − (nominal + 9 mm)	+13.3 [+12.0, +14.7]	+13.6 [+12.2, +15.0]
combined − combined_no_phase	-0.2 [-0.6, +0.2]	-0.1 [-0.3, +0.1]

- Extrapolation: combined − geometry -1.8 [-3.1, -0.7]; − (geometry + const) -12.6 [-16.0, -9.3]
- Trained combined is 0.4 pp below its own untrained (= geometry) version
- Residual RL without geometry learns a state-dependent correction a constant cannot match (+13 pp)
- Explicit contact-phase input: no effect

Why combined does not help

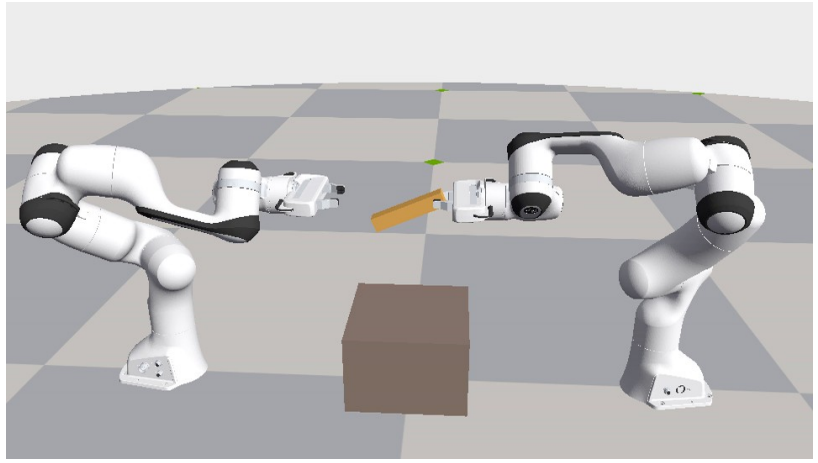
same mechanism in v1 and v2



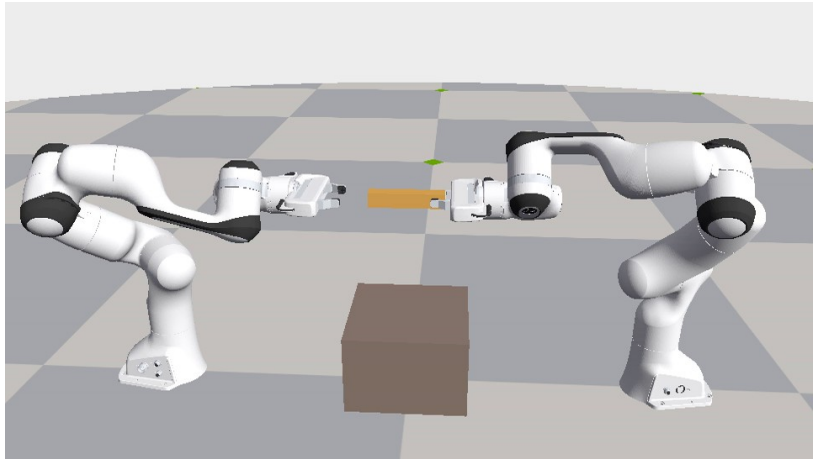
- Learned combined residual ≈ constant −1… −2.4 mm in x (spread ≤ 0.6 mm, slope ≈ 0)
- 92–100 % of combined failures: object shifted > 2 cm toward the receiver (beyond the 1.5 cm bound)
- Those cases need a positive offset; the +10 mm constant solves 97 % of extrapolation cases
- Inside the training range geometry already succeeds → no learning signal for the useful direction
- Residual-only policies see failures in training and learn ~10–13 % offset tracking

Demo (10 provenance-linked videos in the release)

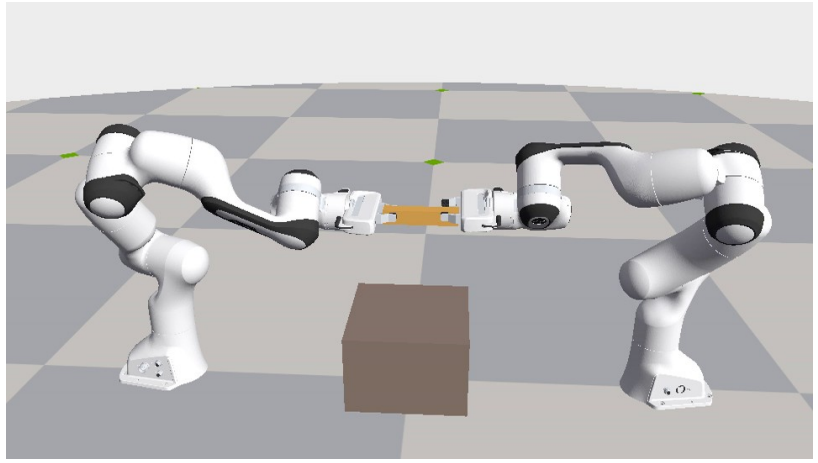
every single-env replay matched its evaluation record



nominal: timeout (joint −0.30 s, ±2 cm)



geometry 1.5 cm: success (same case)



combined: palm collision at ±3 cm

- Matched comparison, release-gate rescue, alignment-bound example, trained success/failure, residual policy
- Selected by a rule frozen before final outcomes (videos.json lists rule, hashes, replay outcome)

Conclusions and limitations

- Strong geometry is sufficient for this handover; alignment does most of the work, gating the rest
- Residual RL on top of geometry adds no robust value and loses to a development-tuned constant offset
- Residual RL alone helps a lot over nominal (+17 pp) but stays far below geometry
- Limitations: simulation only, privileged state, translation-only perturbations, 3 seeds,
  - v2 designed after v1 diagnostics, saturated training range / validation, one reward design
- Everything public: tags experiment-v2 + release v2.0.0 (records, checkpoints, logs, videos, report)